

Balanced Multi-Task Learning for MedMNIST

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Abstract—This paper presents a multi-task learning approach for medical image classification across 11 MedMNIST datasets. The study compares a ResNet-18 baseline, independently trained models, and a shared backbone ConvNeXt-Tiny model with per-task classification heads. The datasets vary significantly in size, number of classes, image modality, and class imbalance, thus making the harmonic mean of per-task macro-F1 scores the primary evaluation metric. The proposed shared model uses weighted cross-entropy with class weights computed per dataset to address imbalance. Experimental results show that shared multi-task training substantially outperforms independent training. The largest gains were observed on small and imbalanced datasets, where shared representations compensate for limited training data. Ablation experiments with focal loss, dataset embeddings, focused fine-tuning, and ensemble combinations did not improve over the simpler shared baseline. The main limitation is performance on inherently ambiguous tasks such as RetinaMNIST, where ordinal severity grades remain difficult to distinguish even with a shared backbone.

Index Terms—MedMNIST, multi-task training, shared backbone, medical image classification, weighted cross-entropy, macro-F1

I. INTRODUCTION

This paper presents a solution for the Tensor Reloaded Multi-Task MedMNIST competition [1]. The goal of the competition is to classify medical images from several MedMNIST tasks and to produce one valid submission file. The study compares a ResNet-18 baseline [2], a shared multi-task ConvNeXt model [3], and independent EfficientNetV2 models trained for each dataset [4]. The main focus is the effect of these choices on the public score and on the per-task validation results.

Deep learning is widely used in medical image analysis, especially with convolutional neural networks [5]. Medical image classification is often tested on one dataset at a time. This competition uses a different setting. It includes eleven 2D MedMNIST datasets, each with its own number of classes, image type, and data size. Some datasets are RGB, while others are grayscale. Some tasks are binary classification problems, while others have more classes. The number of training images also changes strongly between tasks.

MedMNIST provides a common format for several biomedical image classification datasets [6]. The images used in this work have a resolution of 28 by 28 pixels. This makes the benchmark practical under limited GPU time. It shall be noted that the low image resolution does not remove the main difficulties of the task. The datasets still differ in class balance, number of samples, image modality, and label structure.

The multi-task setting is important because a shared model can learn a common representation for related tasks [7], [8]. At the same time, sharing can also hurt some tasks if the datasets are too different. For this reason, the work compares shared training with independent training. The experiments also test focal loss, which was proposed to reduce the effect of class imbalance during training [9].

The experimental process started from a ResNet-18 baseline. A corrected shared ConvNeXt-Tiny model was then trained with one classification head for each dataset. Additional variants tested focal loss with dataset embeddings and independent EfficientNetV2-S models trained separately for each task. The shared multi-task ConvNeXt model is treated as the main method, while the other runs are used to support the ablation study. The other variants did not improve the score, but they provide useful evidence for the ablation study.

The remainder of the paper is organized as follows. Section II presents related work on MedMNIST and image classification models. Section III describes the datasets and the evaluation protocol. Section IV explains the proposed method. Section V gives the experimental setup. Section VI presents the ablation study. Section VII discusses the results. Section VIII describes the main limitations and possible future work. Section IX concludes the paper.

II. RELATED WORK

Deep learning methods have been widely studied for medical image analysis. Litjens et al. reviewed more than 300 papers and showed that convolutional networks were already used for classification, detection, segmentation, and registration in several medical domains [5]. Shen et al. also discussed how deep models can learn useful image representations for medical tasks, especially when they are trained or fine-tuned on task-specific data [10]. These works show why convolutional models are a natural starting point for medical image classification.

MedMNIST was proposed by Yang et al. as a lightweight benchmark for medical image classification [11]. The first version was designed to make medical image experiments easier to run and compare. MedMNIST v2 later extended the benchmark to more datasets and included both 2D and 3D biomedical image tasks [6]. This benchmark is related to the present work because the competition uses multiple 2D MedMNIST datasets in a single evaluation setting.

The baseline architecture used in this work is based on ResNet. He et al. introduced residual learning to make deep

convolutional networks easier to optimize [2]. Their method uses skip connections, so a block learns a residual function instead of a full transformation. Later, Liu et al. proposed ConvNeXt, a modern convolutional architecture that updates standard ConvNets using design choices inspired by recent vision models [3]. EfficientNetV2, proposed by Tan and Le, introduced a family of models focused on faster training and better scaling [4]. These three works motivate the model choices compared in the experiments.

Multi-task learning was introduced by Caruana as a way to improve generalization by learning several related tasks together [7]. In this setting, tasks can share internal representations instead of being trained fully separately. Ruder later summarized multi-task learning in deep neural networks and discussed common designs such as hard parameter sharing, where most layers are shared and each task has its own output head [8]. This is directly related to the shared-backbone model used in this project.

Class imbalance has also been studied in deep learning. Lin et al. proposed focal loss for dense object detection, where many examples are easy negatives [9]. The loss reduces the effect of easy samples and gives more focus to harder samples. For a true-class probability p_t , focal loss is defined as

$$FL(p_t) = -(1 - p_t)^\gamma \log(p_t), \quad (1)$$

where γ controls the down-weighting effect. Although focal loss was introduced for object detection, it is also relevant to classification problems with strong imbalance.

Evaluation metrics for imbalanced classification were studied by Sokolova and Lapalme [12]. They discussed measures based on precision, recall, and F-score, and showed that different metrics can lead to different conclusions. This is important for MedMNIST because several tasks are imbalanced. Macro-F1 is useful in this setting because it gives the same importance to each class, instead of being dominated by frequent classes.

III. DATASET AND EVALUATION PROTOCOL

A. Dataset

The competition is based on eleven 2D datasets from MedMNIST. Each dataset is stored as a NumPy archive with training, validation, and test images. The training and validation labels are available. The test labels are hidden and are evaluated by the Kaggle leaderboard. All images have a resolution of 28×28 pixels. RGB datasets have three channels, while grayscale datasets have one channel.

Table I summarizes the datasets used in this work. The tasks have different sizes, numbers of classes, and imbalance ratios. This is important because the final result depends on all tasks, not only on the largest datasets. RetinaMNIST is marked as ordinal in the MedMNIST metadata, but it is handled as a single-label classification problem in the submitted predictions.

TABLE I
DATASET SUMMARY USED IN THE COMPETITION.

Dataset	Classes	Train	Test	Imb.
PathMNIST	9	89,996	7,180	1.63
DermaMNIST	7	7,007	2,005	58.66
OCTMNIST	4	97,477	1,000	5.93
PneumoniaMNIST	2	4,708	624	2.87
RetinaMNIST	5	1,080	400	7.36
BreastMNIST	2	546	156	2.71
BloodMNIST	8	11,959	3,421	2.74
TissueMNIST	8	165,466	47,280	9.04
OrganAMNIST	11	34,581	17,778	4.54
OrganCMNIST	11	13,000	8,268	5.01
OrganSMNIST	11	13,940	8,829	5.64
Total	–	439,760	96,941	–

B. Data Processing

All models receive three-channel inputs. For RGB datasets, the original image is used directly. For grayscale datasets, the single channel is copied three times. This allows the same ImageNet-pretrained backbones to be used for all datasets. The main experiments keep the original 28×28 resolution.

The training transforms include horizontal flipping and small rotations. Vertical flipping is not used for OCTMNIST and the organ datasets, because the anatomical orientation can carry useful information. Validation and test images use deterministic transforms only. Pixel values are normalized with ImageNet mean and standard deviation, since the backbones are initialized from ImageNet-pretrained models.

C. Submission Format

A valid submission contains one prediction for every test image from every task. The required columns are

$$\text{id, id_image_in_task, task_name, label.} \quad (2)$$

The row order follows the dataset order shown in Table I. The final submission must contain exactly 96,941 rows. Before every upload, the submission file is checked for column order, row count, missing labels, monotonic global identifiers, and the expected number of rows for each dataset.

D. Validation Metrics

The local validation score is computed separately for each dataset. Since several datasets are imbalanced, macro-F1 is used instead of accuracy. Macro-F1 gives the same weight to each class, so frequent classes do not dominate the score. This is important for datasets such as DermaMNIST, RetinaMNIST, and TissueMNIST, where class imbalance or small validation size can strongly affect the result.

To compare full multi-task runs, the harmonic mean of the eleven dataset macro-F1 scores is used as the local global score:

$$H = \frac{N}{\sum_{i=1}^N \frac{1}{F1_i}}, \quad (3)$$

where $N = 11$ and $F1_i$ is the validation macro-F1 of dataset i . This metric gives more importance to weak tasks than

the arithmetic mean. As a result, improving only the easiest datasets is not enough. The model must also improve the tasks with lower macro-F1.

It shall be noted that the local harmonic mean is used only for model selection and ablation. The final benchmark performance is the public Kaggle leaderboard score.

IV. PROPOSED METHOD

The proposed method is a shared multi-task convolutional model. It uses one common image backbone and one classification head for each MedMNIST dataset. This design was chosen because all tasks are medical image classification tasks, but each dataset has its own number of classes. The shared backbone learns general image features, while the task-specific heads map these features to the labels of each dataset.

A. Input Representation

All input images are converted to three channels. RGB images are used directly. For grayscale datasets, the single channel is repeated three times. This makes it possible to use the same ImageNet-pretrained backbone for all tasks. The main model uses the original MedMNIST resolution of 28×28 pixels. No task-specific image size is used.

B. Shared Backbone

The main model uses ConvNeXt-Tiny as the shared backbone. The final classifier of the original model is removed, so the backbone only produces feature vectors. The first convolutional layer is adapted for 28×28 inputs by using a smaller stride. This avoids reducing the spatial resolution too early.

Let x be an input image and let $B(\cdot)$ be the shared backbone. The feature vector is

$$z = B(x). \quad (4)$$

The same function B is used for all datasets.

C. Task-Specific Heads

Each dataset has its own classification head. If dataset t has C_t classes, its head outputs C_t logits. This avoids using one large output space for all tasks. It also allows each task to keep its own label set.

For task t , the prediction logits are

$$y_t = H_t(z), \quad (5)$$

where H_t is the classification head for task t . During training, each sample contributes only through the head associated with its dataset. During inference, the corresponding head is selected based on the task name.

D. Loss Function

The main model uses weighted cross-entropy. Class weights are computed from the training labels of each dataset. This gives more importance to rare classes and reduces the effect of class imbalance. For dataset t , the loss is computed only over the classes of that dataset:

$$\mathcal{L}_t = CE_w(y_t, c_t), \quad (6)$$

where c_t is the ground-truth label and CE_w is weighted cross-entropy. The total loss for a mini-batch is the average of the task losses present in that batch:

$$\mathcal{L} = \frac{1}{|\mathcal{T}_b|} \sum_{t \in \mathcal{T}_b} \mathcal{L}_t, \quad (7)$$

where $\mathcal{T}_b$ is the set of tasks present in the mini-batch and $|\mathcal{T}_b|$ is its size.

E. Model Variants

The shared ConvNeXt-Tiny model is the main method studied in this paper. Several additional models are used only for comparison in the ablation study. These include a ResNet-18 baseline, a focal-loss variant with dataset embeddings, and independent EfficientNetV2-S models trained separately for each dataset. These variants are used to study the effect of architecture, loss function, and parameter sharing.

V. EXPERIMENTAL SETUP

All experiments were run in the Kaggle environment. The recorded runs used Python with PyTorch 2.10.0, CUDA 12.8, and the `timm` model library. The available GPU was an NVIDIA Tesla T4 with 15.6 GB of memory. The data were loaded from the Kaggle competition input directory, while checkpoints, result files, and submissions were written to the Kaggle working directory.

A fixed random seed of 42 was used in the main runs. The same dataset order and submission validation procedure were used for all full submissions. Each submitted file was checked for column order, row count, missing labels, monotonic identifiers, and the expected number of samples for each task.

A. Training Protocol

The experiments used ImageNet-pretrained convolutional backbones. The baseline and independent models were trained separately for each dataset. The main model used a shared backbone and one head per dataset. For the shared model, each mini-batch could contain samples from different tasks, and the loss was averaged over the tasks present in the batch.

The optimizer was AdamW in the main neural network experiments. Learning rates, batch sizes, and stopping rules were changed empirically depending on the run. Cosine learning-rate scheduling was used in the full-dataset runs. The best checkpoint was selected using the validation metric of the corresponding run. For shared multi-task runs, this was the validation harmonic mean. For independent per-dataset runs, this was the validation macro-F1 of each dataset.

B. Experimental Runs

The experiments were organized around one baseline and several controlled variants. Some tasks, like the baseline and final submission were trained on all the datasets, while other runs tested independent models, focal loss, dataset embeddings, and focused tuning for the smallest datasets.

TABLE II
SUMMARY OF THE EXPERIMENTAL RUNS.

Run	Model	Training	Image	Submitted
Baseline	ResNet-18	Independent	28	Yes
EfficientNetV2-S 64	EfficientNetV2-S	Independent	64	Yes
Shared ConvNeXt	ConvNeXt-Tiny	Multi-task	28	Yes
Focal Embedding	ConvNeXt-Tiny	Multi-task	28	No
EfficientNetV2-S 28	EfficientNetV2-S	Independent	28	Yes
Small Dataset Focus	ConvNeXt-Tiny	Retina/Breast	28	Partial

The EfficientNetV2-S 64 run tested whether a larger input size and independent training could improve the baseline. The shared ConvNeXt run was the main corrected multi-task model. The focal embedding run added focal loss for high-imbalance datasets and a learnable dataset embedding. It was not submitted because the validation behavior did not show an improvement. The EfficientNetV2-S 28 run tested independent per-dataset training at the original image size. The small dataset focused run trained only on RetinaMNIST and BreastMNIST and while it showed marginal improvements for these sets, these were not enough to increase the overall target metric of the public submission.

C. Run Configurations

The main hyperparameters are shown in Table III. The ConvNeXt runs used `convnext_tiny`. The EfficientNetV2-S runs used `tf_efficientnetv2_s`.

TABLE III
MAIN HYPERPARAMETERS FOR THE EXPERIMENTAL RUNS.

Run	Batch	Epochs	LR	WD	Early Stop
EfficientNetV2-S 64	128	5+25	$1e-4/1e-3$	$1e-4$	7
Shared ConvNeXt	512	25	$1e-4$	$5e-2$	5
Focal Embedding	512	25	$1e-4$	$5e-2$	5
EfficientNetV2-S 28	128	3+20	$1e-4/1e-3$	$1e-4$	7
Small Dataset Focus	32	60	$5e-5$	$5e-3$	20

For the independent EfficientNetV2-S runs, training was split into two phases. In the first phase, the backbone was frozen and only the classification head was trained. In the second phase, the full model was fine-tuned with a lower learning rate for the backbone and a higher learning rate for the head. For the shared ConvNeXt runs, the full model was trained with one optimizer and a cosine learning-rate schedule.

D. Augmentation and Inference

The training augmentations used horizontal flips, small rotations, and medium augmentation in the EfficientNetV2-S 64 run. Vertical flips were avoided for OCTMNIST and the organ datasets. The main shared ConvNeXt run did not use test-time augmentation. The EfficientNetV2-S 64 run used test-time augmentation where allowed.

For each run, predictions were generated from the best saved checkpoint, not from the last epoch. Full submissions contained all eleven datasets and 96,941 predictions. The small dataset focused run produced only RetinaMNIST and BreastMNIST predictions and was used as a focused experiment, not as a complete competition submission.

E. Recorded Artifacts

Each Kaggle run exported a submission file, a run configuration file, and a results file. The results files stored the validation macro-F1 for each dataset and the settings needed for later comparison. These artifacts were used to build the ablation tables and to compare local validation behavior with public leaderboard scores.

VI. ABLATION STUDY

The ablation study evaluates the main choices that affected the system. The experiments are grouped by the factor being changed: architecture, parameter sharing, input resolution, loss design, focused tuning for small datasets, and computational cost. Full runs are compared with the local harmonic mean over all eleven datasets. Focused runs are compared only on the datasets used in that experiment.

It shall be noted that not all ablations improved the validation or public score. These runs are still included because they show which directions were not useful under the available compute and validation setting.

A. Architecture and Parameter Sharing

The first ablation compares independent training with shared multi-task training. The ResNet-18 model is used as the initial reference. EfficientNetV2-S is used to test whether a stronger independent model improves the result. ConvNeXt-Tiny is used to test whether sharing one backbone across all datasets is more effective than training one model per dataset.

TABLE IV
ARCHITECTURE AND PARAMETER SHARING ABLATION.

Model	Training	Local HM	Public
ResNet-18	Independent	0.6490	0.599
EfficientNetV2-S, 28	Independent	0.6572	–
EfficientNetV2-S, 64	Independent	0.6665	–
ConvNeXt-Tiny	Shared	0.7835	0.725

The shared ConvNeXt-Tiny model gives the highest local harmonic mean. The public score also increases compared with the ResNet-18 baseline. The EfficientNetV2-S independent runs remain close to the baseline, even though they use a stronger backbone. Increasing the input size from 28×28 to 64×64 gives only a small gain for EfficientNetV2-S, from 0.6572 to 0.6665.

B. Per-Dataset Effects

The harmonic mean is sensitive to weak tasks, so the global score alone is not enough. Table V compares the per-dataset validation macro-F1 of the main models.

TABLE V
PER-DATASET VALIDATION MACRO-F1 FOR THE MAIN MODELS.

Dataset	EffV2S-64	EffV2S-28	Shared ConvNeXt
PathMNIST	0.9917	0.9721	0.9750
DermaMNIST	0.3691	0.3717	0.5804
OCTMNIST	0.9122	0.8459	0.8541
PneumoniaMNIST	0.9419	0.9149	0.9824
RetinaMNIST	0.2931	0.3511	0.4816
BreastMNIST	0.7292	0.5334	0.8655
BloodMNIST	0.9424	0.9250	0.9719
TissueMNIST	0.6423	0.5889	0.6062
OrganAMNIST	0.9865	0.9798	0.9881
OrganCMNIST	0.9730	0.9745	0.9821
OrganSMNIST	0.8441	0.8366	0.8609
Global HM	0.6665	0.6572	0.7835

The shared model improves several weak datasets, especially DermaMNIST, RetinaMNIST, and BreastMNIST. These gains are important because weak tasks have a large effect on the harmonic mean. The independent EfficientNetV2-S models obtain high scores on some datasets, but they do not improve the weakest tasks enough.

C. Loss Function and Dataset Embedding

Class imbalance was one of the main issues observed during validation. A shared ConvNeXt variant was therefore tested with focal loss on high-imbalance datasets and weighted cross-entropy on the remaining datasets. A learnable dataset embedding was also added before the task-specific heads.

This variant was not kept as the main model because it did not show better validation behavior than the weighted cross-entropy shared ConvNeXt model. Since no complete leaderboard submission and no finalized result table were produced for this run, it is not included as a numerical comparison. The experiment is still relevant because it tested whether a more complex loss setup and explicit task information helped the imbalanced datasets. In this case, the simpler weighted cross-entropy setup was kept as the main model.

D. Small Dataset Focus

The next ablation focuses on RetinaMNIST and BreastMNIST. These are the two smallest datasets, and RetinaMNIST was one of the weakest tasks in the main shared model. A focused model was trained only on these two datasets, with a smaller batch size, a lower learning rate, stronger regularization, and longer training.

TABLE VI
FOCUSED TUNING FOR THE TWO SMALLEST DATASETS.

Dataset	Shared ConvNeXt	Focused Model
RetinaMNIST	0.4816	0.4961
BreastMNIST	0.8655	0.8697
Partial HM	0.6170	0.6318

The focused run gives a small validation improvement on both datasets. The harmonic mean in Table VI is computed only over RetinaMNIST and BreastMNIST, so it is not directly comparable with the global harmonic mean from the full

eleven-dataset runs. The gain is larger for RetinaMNIST than for BreastMNIST, which was already strong in the shared model.

E. Computational Cost

For this study, runtime is also taken into account, since we were limited by the Kaggle compute environment. Table VII reports the recorded runtime for selected runs. The runtimes are provided as additional information and are not used as the main selection criterion, since each experiment used different model architecture and hyperparameters.

TABLE VII
RECORDED RUNTIME FOR SELECTED RUNS.

Run	Scope	Runtime
EfficientNetV2-S, 64	Full dataset	5.3 h
EfficientNetV2-S, 28	Full dataset	1.8 h
ConvNeXt-Tiny Shared	Full dataset	13.0 h
Small Dataset Focus	Retina/Breast	8.2 min

The shared ConvNeXt-Tiny run was more expensive than the independent EfficientNetV2-S runs. However, the difference in validation and public performance was also much larger. The faster EfficientNetV2-S runs were useful for comparison, but their lower runtime did not compensate for the weaker harmonic mean. The small-dataset run was cheap to train, but it produced only a limited improvement.

F. Ensemble Study

After tuning a model on the two smallest datasets (RetinaMNIST and BreastMNIST), it was used in conjunction with the common backbone model to see if an ensemble of a specialized model and a common one can lead to better results, but this was not the case.

TABLE VIII
ENSEMBLE VS. FINAL MODEL ON THE PUBLIC LEADERBOARD.

	Shared ConvNeXt	Ensemble
Public score (HM F1)	0.725	0.72

VII. RESULTS AND DISCUSSION

This section reports the final validation and leaderboard results. The main comparison is made between the initial ResNet-18 baseline, the independent EfficientNetV2-S runs, and the shared ConvNeXt-Tiny model. The local score is the harmonic mean of the eleven validation macro-F1 scores, which can be observed in Table IX. The public score is the Kaggle leaderboard score computed on the hidden test labels.

The shared ConvNeXt-Tiny model obtained the best local harmonic mean, with a harmonic mean of 0.783 and public score of 0.725. This was a clear improvement over the ResNet-18 baseline, which obtained 0.649 locally and 0.599 on the public leaderboard respectively. This proves that the local validation protocol was useful for model selection, even though the validation score and the public score are not identical.

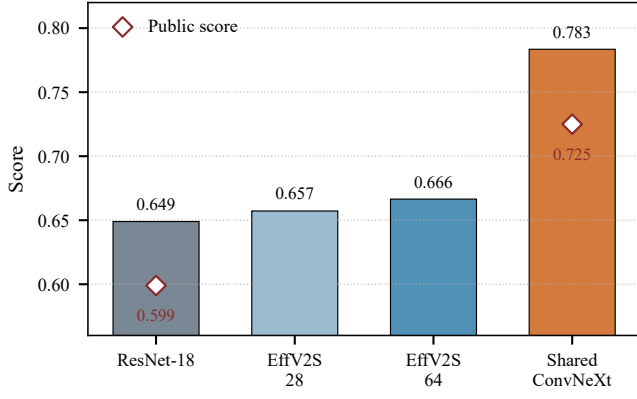


Fig. 1. Global comparison between the main experimental runs.

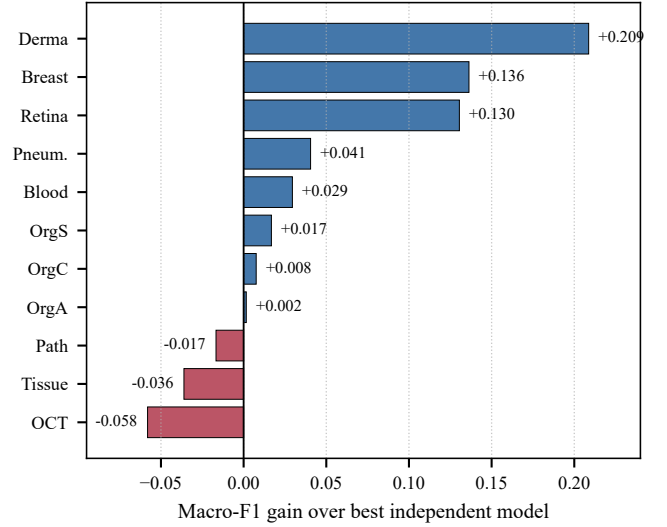


Fig. 2. Per-dataset macro-F1 gain of the shared model over the best independent run for each dataset.

TABLE IX
PER-DATASET MACRO F1 SCORES ON THE TEST SET.

Dataset	Train samples	Macro F1
PathMNIST	89,996	0.8803
DermaMNIST	7,007	0.6208
OCTMNIST	97,477	0.8050
PneumoniaMNIST	4,708	0.8254
RetinaMNIST	1,080	0.3718
BreastMNIST	546	0.7949
BloodMNIST	11,959	0.9702
TissueMNIST	165,466	0.6104
OrganAMNIST	34,581	0.9400
OrganCMNIST	13,000	0.9143
OrganSMNIST	13,940	0.7698
Harmonic mean		0.7202

Figure 2 shows where the shared model helped most. The largest gains were obtained on DermaMNIST, BreastMNIST, and RetinaMNIST. These datasets were important because they had lower validation scores in the independent runs and therefore had a strong effect on the harmonic mean. The shared model improved DermaMNIST by 0.209 macro-F1, BreastMNIST by 0.136, and RetinaMNIST by 0.130 compared with the best independent EfficientNetV2-S result for each dataset.

Figure 3 shows that dataset size alone does not explain the result. It can be quickly observed that for sets which are relatively balanced and have enough training instances, such as BloodMNIST, the model achieves high accuracy 5. The same is true for OrganAMNIST and OrganCMnist, which can also be attributed to anatomically separable organ, as it can be seen in Figure IX. At the opposite extreme, RetinaMNIST yields the lowest score as it can be seen in Figure 6, a result attributed to ambiguity of its five ordinal severity grades 8.

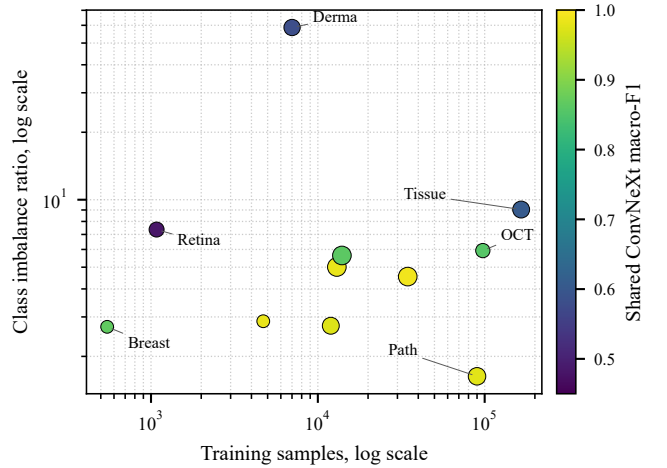


Fig. 3. Dataset difficulty map using the number of training samples, class imbalance ratio, and shared ConvNeXt validation macro-F1. Both axes use a logarithmic scale.

VIII. LIMITATIONS AND FUTURE DIRECTIONS

One of the main limitations encountered during this study were computational and time constraints. Figure 4 shows

the time each run took. The shared ConvNeXt-Tiny model required about 13 hours in the Kaggle environment, while the independent EfficientNetV2-S runs were faster. The cost is justified as the performance gain is proportional.

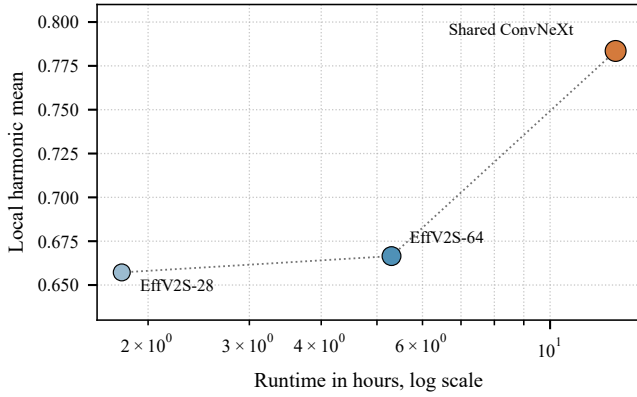


Fig. 4. Runtime and validation harmonic mean for the main full-dataset runs with recorded execution times.

Another limitation is represented by the complexity of each task. The macro-F1 score computed per dataset reveals that some datasets are considerably harder to learn, and even though attempts were made to isolate and do finer tuning on these particular sets, the gains were not significant. Finally, training was mostly done on the 28×28 datasets. While it was shown that increasing the input resolution can give some improvements, this direction was not thoroughly investigated.

As such, future directions could focus on allocating more training time and computational power, increasing input size and fine tuning individual models on the weakest performing datasets, like RetinaMNIST, DermaMNIST and TissueMNIST, either through a hyperparameter gridsearch or sweep.

IX. CONCLUSION

This paper presented a multi-task learning approach for the Tensor Reloaded MedMNIST competition, comparing a ResNet-18 baseline, independently trained models, and a shared backbone model across eleven MedMNIST datasets.

The results show that shared multi-task training with a common backbone and per-task classification heads significantly outperforms independent training. The best results were obtained on the shared ConvNeXt-Tiny model with a score of 0.725. The improvement was most pronounced on the weakest and most imbalanced datasets: DermaMNIST, BreastMNIST, and RetinaMNIST; which had the greatest influence on the harmonic mean metric. This confirms that shared representations can benefit tasks with limited training data, where independent models struggle to generalize. The ablation study further showed that more complex alternatives, including focal loss with dataset embeddings, focused fine-tuning on small datasets, and ensemble combinations, did not consistently improve over the simpler weighted cross-entropy shared model.

The main remaining challenge is performance on inherently difficult datasets such as RetinaMNIST, where the ambiguity between ordinal severity grades limits classification accuracy regardless of architecture.

Future work should explore higher input resolutions, task-adaptive loss weighting, and more extensive hyperparameter search for the weakest-performing tasks.

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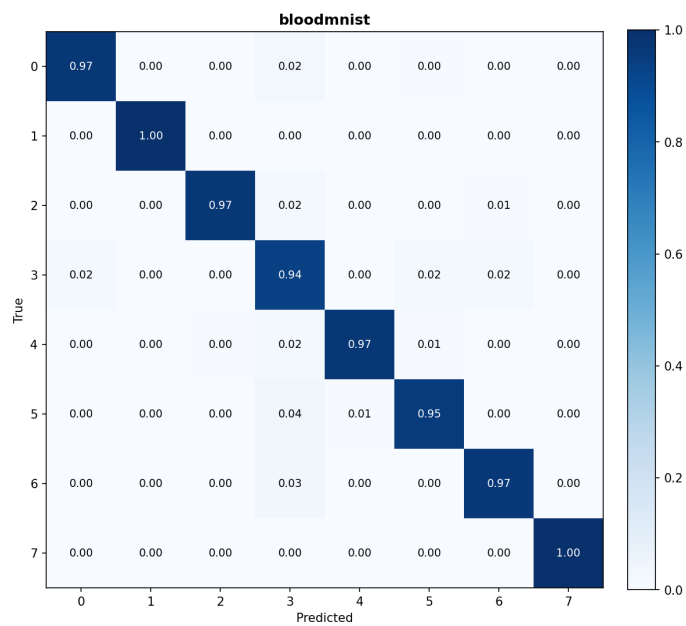


Fig. 5. BloodMNIST confusion matrix

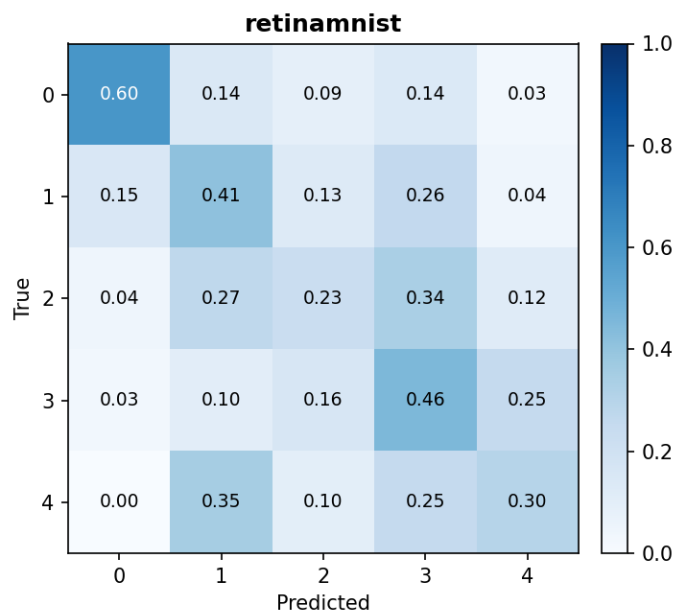


Fig. 6. RetinaMNIST confusion matrix

OrganAMNIST Comparison

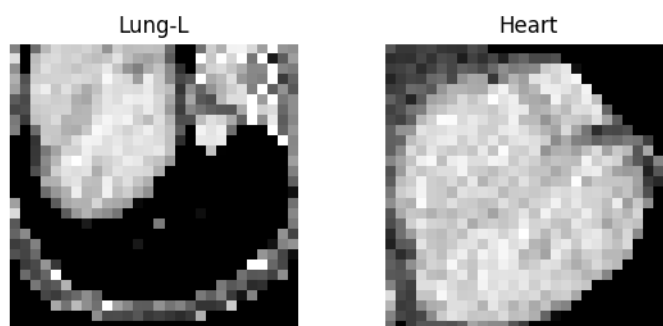


Fig. 7. Two OrganCMNIST classes

RetinaMNIST - severity grades

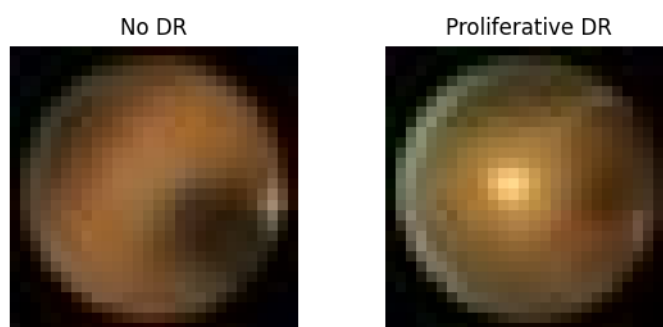


Fig. 8. Two RetinaMNIST classes